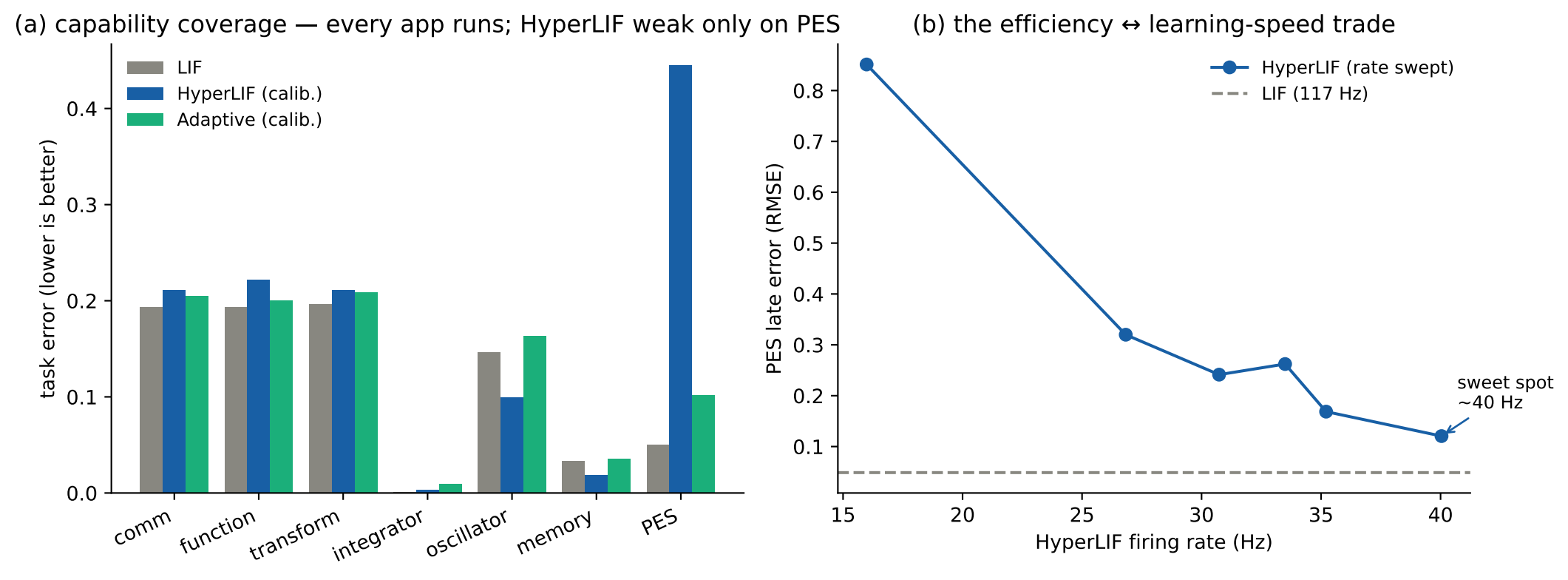
(a) capability coverage — every app runs; HyperLIF weak only on PES
LIF
HyperLIF (calib.)
Adaptive (calib.)
task error (lower is better)
0.0
0.1
0.2
0.3
0.4
comm
function
transform
integrator
oscillator
memory
PES
(b) the efficiency ↔ learning-speed trade
HyperLIF (rate swept)
LIF (117 Hz)
PES late error (RMSE)
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
sweet spot
~40 Hz
15
20
25
30
35
40
HyperLIF firing rate (Hz)